

APPLICATION OVERVIEW DOCUMENT

Hertz Fleet Operating System

Congratulations on completing your application definition! This document presents a summary overview of your application as defined in the Pega platform. It encapsulates the key elements of your application's design, highlighting its core functionalities and overall structure.

Please note that this is a high-level view and does not include granular details like data models or sample data. It's an ideal resource for a quick reference, facilitating discussions among team members and stakeholders, and for guiding strategic decisions in your application development process.

Blueprint ID: BP-2028128

Application Context

This application revolutionizes car rental operations by establishing a comprehensive fleet operating system. It addresses the immense complexity of managing over half a million vehicles across thousands of global locations, ensuring the right car is available at the right place and time for customers. The system integrates fleet and workforce management, orchestrating critical processes like vehicle cleaning, maintenance, and movement. Key features include a "fleet control tower" for a bird's eye view of operations, AI agents for proactive issue identification based on external factors (weather, flights, demand) and internal staffing challenges, and a lightweight mobile application providing "next best action" intelligence to on-lot transporters and employees. It automates task generation, allows human-in-the-loop adjustments, and enables employees to accept or reject tasks via a mobile app. This intelligent, app-driven ecosystem replaces manual dispatching, leading to significant improvements in operational efficiency, minimized maintenance costs, enhanced car availability, optimized labor deployment and a superior customer experience.

Application Architecture

Users access the application through defined channels, assume specific roles, and trigger workflows that manage how work progresses. Those workflows operate on shared Data Objects to ensure the system stays consistent and up to date.

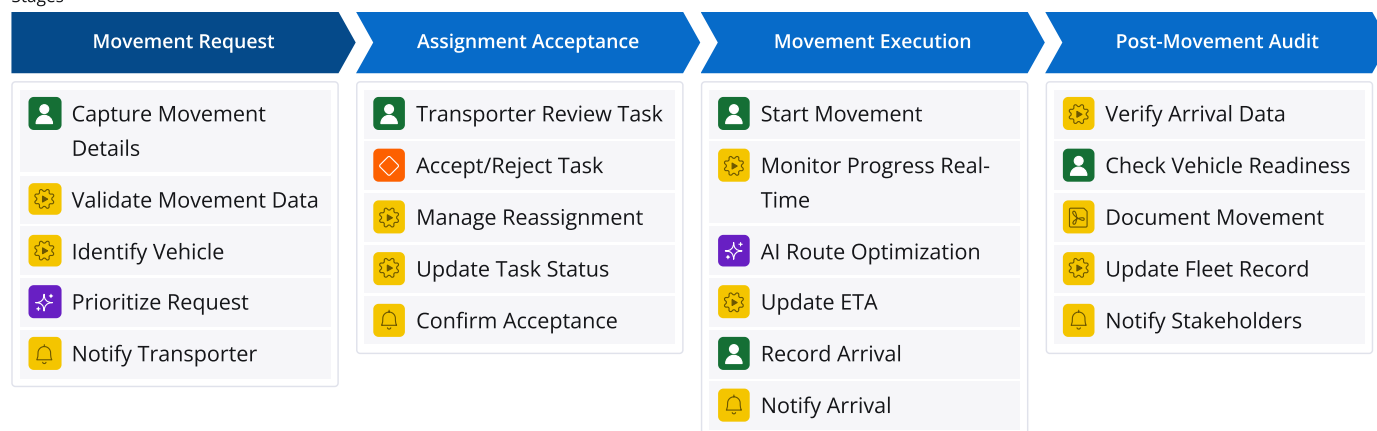
6**PERSONAS AND CHANNELS****Customer****Transporter****Vehicle Service Associate****Front Desk Staff****Operations Manager****AI Agent Specialist****5****WORKFLOWS****Vehicle Movement****Vehicle Cleaning****Fleet Maintenance****Staffing Adjustment****Operational Incident****10****DATA OBJECTS****System of record: Pega(Local)
Vehicle****System of record: Pega(Local)
Location****System of record: Pega(Local)
Customer****System of record: Pega(Local)
Employee****System of record: Pega(Local)
Task****System of record: Pega(Local)
Weather Event****System of record: Pega(Local)
Flight****System of record: Pega(Local)
Returns Queue****System of record: Pega(Local)
Dirty Car Backlog****System of record: Pega(Local)
Overtime**

Workflows (Case Types)

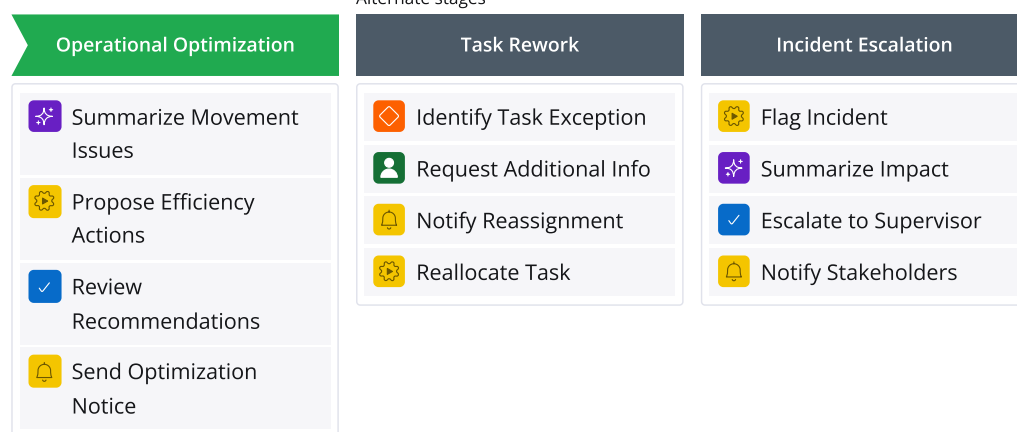
Vehicle Movement

Orchestrates the end-to-end process of relocating vehicles across multiple sites and airport lots to ensure optimal availability, using real-time intelligence and AI-driven recommendations. Enables transporters to accept and execute movement tasks, reducing delays and improving customer fulfillment.

Stages



Alternate stages



Case Data Model

Field Name	Description	Type	Primary ⓘ
Vehicle movement n...	Descriptive name given to this specific vehicle movement task or instance.	Text	✓
Scheduled move time	The date and time the vehicle is scheduled to be moved.	DateTime	✓
Task acceptance	Indicates whether the assigned employee has accepted or rejected the movement task in thei...	Text	✓
Vehicle	The specific car selected for movement during this process.	(Data)	
Origin location	The Hertz facility, branch, or airport from which the vehicle is being relocated.	(Data)	
Destination location	The Hertz facility, branch, or airport where the vehicle is to be delivered or stationed after mo...	(Data)	
Transporter	The employee designated to perform the vehicle movement, such as a transporter or vehicle ...	(Data)	
AI movement recom...	Explanation or rationale generated by the AI system for why this particular movement was rec...	Text	
Triggering event	Key event or factor that prompted the vehicle movement, such as increased demand, flight ar...	Text	
Task completion time	The actual date and time when the vehicle movement task was finished.	DateTime	
Operational status in...	Status of the movement task, reflecting operational health and urgency derived from the cont...	Text	
Notes	Additional comments, circumstances, or special instructions related to the vehicle movement ...	Text	
Associated task	The operational task record associated with this movement for tracking acceptance, progress,...	(Data)	
Weather event impact	Identifies if a current weather event at the origin or destination location is influencing the sch...	(Data)	

Field Name	Description	Type	Primary
Linked flight	Flight record influencing this vehicle movement, typically for airport-based logistics.	(Data)	
Customer reservatio...	Indicates if this movement is directly supporting a real-time customer reservation or assignm...	Boolean	
Dirty car backlog im...	Indicates whether dirty car backlog at either location affected the movement task prioritization.	(Data)	
Returns queue positi...	Position in the returns queue at the origin branch when the movement was triggered, if applic...	(Data)	
Overtime required	Indicates whether overtime approval was required to execute this movement, due to surges, ...	Boolean	
Case Label	Label for this individual case instance	Text	✓
Case ID	Unique identifier for this individual case instance	Text	✓
Work Status	Work status of the case	Text	✓
Urgency	Priority/urgency of the case	Decimal	
Description	Description for this individual case instance	Text	✓
Created by	User that created the case	User Ref...	
Created date/time	Date/time case was created	DateTime	
Resolved by	User that resolved the case	User Ref...	
Resolved date/time	Date/time case was resolved	DateTime	
Vehicle to be Relocat...	The specific vehicle that is the subject of this movement task.	(Data)	
Vehicle Origin Locati...	The starting location from which the vehicle movement is initiated.	(Data)	
Vehicle Destination L...	The target location where the vehicle is to be delivered.	(Data)	
Assigned Transporter	The employee assigned to accept and execute this vehicle movement task.	(Data)	
Associated Moveme...	The specific operational task record detailing the vehicle relocation, including its status and pr...	(Data)	
Impacting Weather E...	Records of weather-related disruptions that may affect the vehicle movement schedule or rou...	(Data)	
Related Airport Flights	Flight information relevant to anticipating vehicle demand or returns at airport locations, influ...	(Data)	
Vehicle from Returns...	A vehicle that has been returned by a customer and is now being moved from the returns qu...	(Data)	
Vehicle from Dirty C...	A vehicle requiring cleaning or maintenance that is being moved from the dirty car backlog to ...	(Data)	

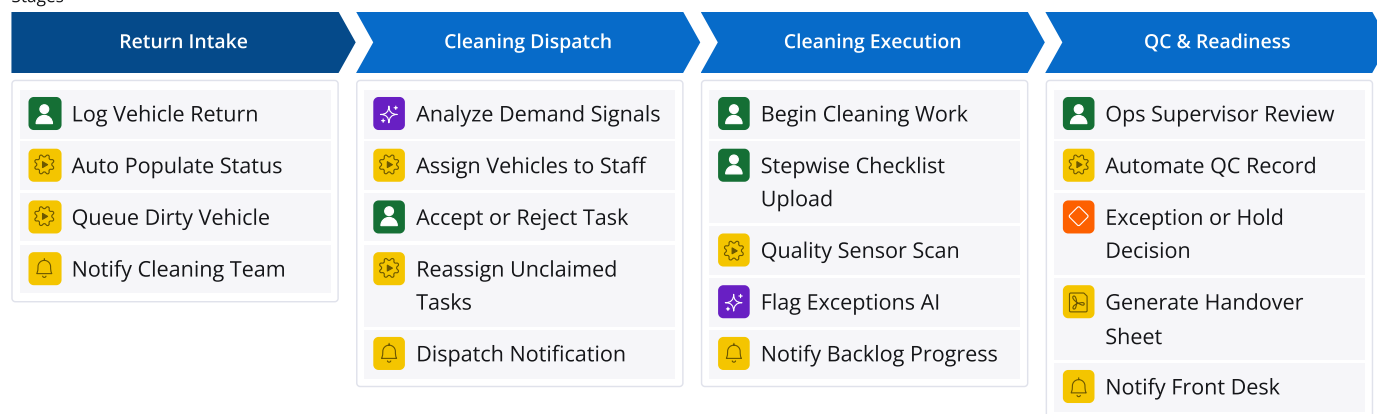
Business Rules

Name	Description
Vehicle Availability O...	Determines optimal vehicle placement and movement priorities based on real-time customer demand, return patterns, ...
Next Best Action Rec...	Synthesizes real-time operational data including vehicle locations, task priorities, workforce availability, and customer de...
Operational Status I...	Determines operational health status (red for backlog, amber for potential issues, green for normal) by evaluating key m...
Task Acceptance Vali...	Validates employee task acceptance or rejection decisions against operational constraints, workforce capacity, and busin...
Vehicle Flow Optimiz...	Orchestrates the sequential movement of vehicles through operational stages (returns, cleaning, maintenance, front des...

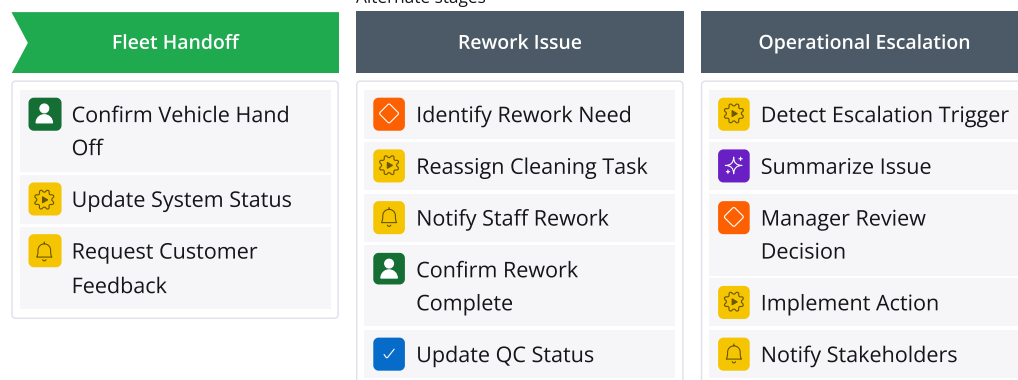
Vehicle Cleaning

Manages the workflow for cleaning and preparing vehicles between rentals, tracking backlog and real-time progress. Automates task assignments to employees, allowing for proactive issue identification and ensuring high-quality, ready-to-rent cars.

Stages



Alternate stages



Case Data Model

Field Name	Description	Type	Primary ⓘ
Vehicle cleaning name	Business-friendly name of the vehicle cleaning instance, uniquely identifying each cleaning a...	Text	✓
Vehicle	The specific vehicle undergoing cleaning, selected from the managed rental fleet.	(Data)	✓
Location	Branch, depot, or airport facility where the cleaning occurs to contextualize logistics and wor...	(Data)	✓
Employee assigned	Workforce member responsible for performing or overseeing this cleaning event.	(Data)	✓
Cleaning start time	Date and time when the vehicle cleaning task is initiated.	DateTime	
Cleaning end time	Date and time when the vehicle cleaning task is completed and vehicle is ready for next oper...	DateTime	
Task status	Current progress indicator for the cleaning operation (e.g., Pending, In Progress, Completed, ...	Text	
Cleaning quality score	Numeric rating reflecting the quality of the vehicle cleaning, often based on predefined inspe...	Integer	
Overtime required	Indicates if overtime was required for cleaning due to high backlog or operational delays.	Boolean	
Task acceptance stat...	Tracks whether the assigned employee accepted, rejected, or requested adjustment for the ...	Text	
Issue detected	Indicates if any issue was detected during cleaning, such as damage or missing equipment.	Boolean	
Issue description	Further explanation provided when an issue is detected during cleaning, supporting later act...	Text	
Backlog status	Current state of the dirty car backlog impacting cleaning prioritization (e.g., High, Moderate, ...	Text	
Returns queue positi...	The vehicle's current place in the incoming returns queue before cleaning assignment to trac...	Integer	
Weather event impact	Associates impactful weather event with cleaning task, used for operational adjustment and ...	(Data)	
Flight impact	Links relevant flight arrival or departure data to cleaning prioritization to anticipate demand ...	(Data)	
Customer	Links the upcoming customer scheduled for the vehicle post-cleaning, supporting satisfactio...	(Data)	

Field Name	Description	Type	Primary
Task	The task record for this cleaning operation, supporting tracking, audit, and integration with b...	(Data)	
Case Label	Label for this individual case instance	Text	✓
Case ID	Unique identifier for this individual case instance	Text	✓
Work Status	Work status of the case	Text	✓
Urgency	Priority/urgency of the case	Decimal	
Description	Description for this individual case instance	Text	✓
Created by	User that created the case	User Refe...	
Created date/time	Date/time case was created	DateTime	
Resolved by	User that resolved the case	User Refe...	
Resolved date/time	Date/time case was resolved	DateTime	
Vehicle to be Cleaned	The specific vehicle that requires cleaning and preparation for the next rental.	(Data)	
Cleaning Location	The facility where the vehicle cleaning and preparation takes place.	(Data)	
Assigned Employee	The employee assigned to perform the vehicle cleaning and preparation tasks.	(Data)	
Cleaning Task	The specific operational task created for cleaning and preparing a vehicle.	(Data)	
Impacting Weather E...	Any weather event that might affect the vehicle cleaning schedule or process.	(Data)	
Vehicle Returns Que...	An entry in the queue for vehicles that have been returned and are awaiting cleaning.	(Data)	
Dirty Car Backlog Ent...	An entry in the backlog of vehicles that require cleaning and maintenance.	(Data)	
Employee Overtime ...	An overtime request related to the increased demand for vehicle cleaning and preparation.	Text	

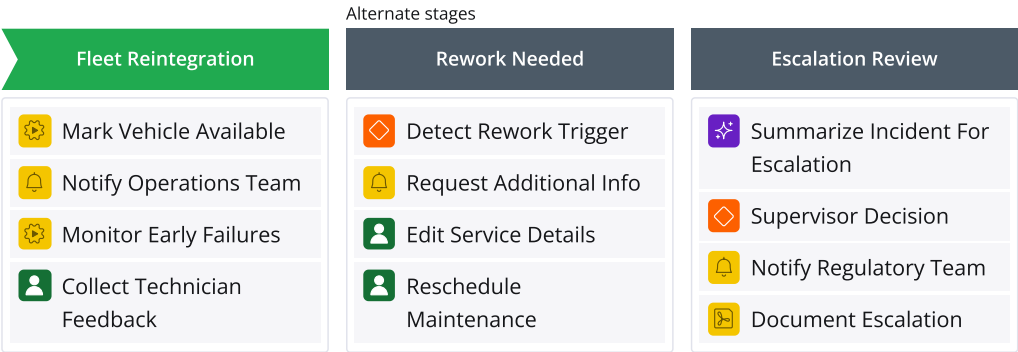
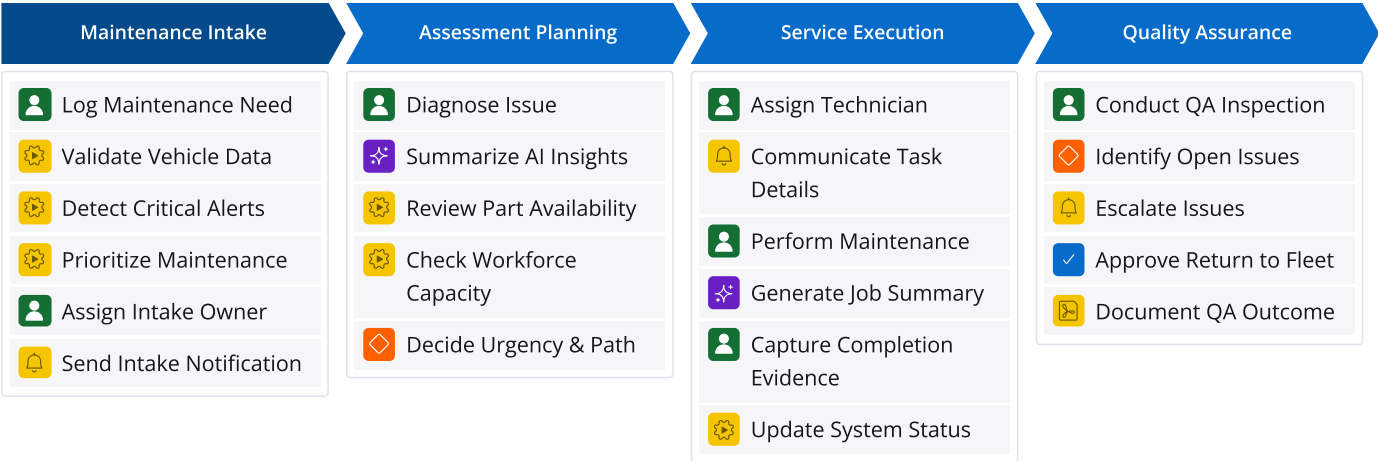
Business Rules

Name	Description
Cleaning Backlog Pri...	Evaluates dirty car backlog severity, upcoming customer reservations, and available workforce capacity to prioritize and a...
Exception or Hold D...	Flag and decide on hold for errors, rework, or damage cases.
Identify Rework Need	AI or supervisor flags vehicle for re-clean after failed QC or sensor alert.

Fleet Maintenance

Coordinates maintenance activities including inspection, servicing, and repair to minimize downtime and ensure vehicles meet operational standards. Integrates AI alerts and workforce adjustments to optimize resource allocation.

Stages



Case Data Model

Field Name	Description	Type	Primary <i>i</i>
Fleet maintenance n...	Business name that uniquely identifies this maintenance case for fleet operations.	Text	✓
Vehicle	The vehicle undergoing maintenance.	(Data)	✓
Maintenance type	Kind of maintenance activity being performed, such as inspection, scheduled servicing, or ...	Text	✓
Maintenance priority	Operational priority level assigned to the maintenance task based on impact and urgency.	Text	✓
Location	The facility or branch where the maintenance is being performed.	(Data)	
Task	Specific operational tasks generated for this maintenance activity, such as inspection or m...	(Data)	
Employee assigned	Primary employee assigned to lead the maintenance.	(Data)	
Requested overtime	Indication if overtime is requested for this maintenance activity.	(Data)	
Proactive maintenanc...	Indication whether this maintenance was triggered by an AI agent's predictive alert.	Boolean	
Weather event impact	Weather event associated with or impacting the maintenance activity, if applicable.	(Data)	
Flight delay impact	Flight disruptions (arriving or departing) that have triggered or influenced maintenance sc...	(Data)	
Inspected at	Date and time when the vehicle was inspected as part of maintenance.	DateTime	
Servicing completed at	Date and time when the maintenance servicing was completed.	DateTime	
Maintenance remarks	Comments added by employees regarding special maintenance circumstances or actions t...	Text	
Customer impact flag	Indicates whether the maintenance activity directly impacts vehicle allocation for a custom...	Boolean	
Returns queue impact	Reference to the returns queue element this vehicle is part of during maintenance process...	(Data)	
Dirty car backlog im...	Reference to the dirty car backlog, highlighting vehicles that must be cleaned post-mainte...	(Data)	

Field Name	Description	Type	Primary
Case Label	Label for this individual case instance	Text	✓
Case ID	Unique identifier for this individual case instance	Text	✓
Work Status	Work status of the case	Text	✓
Urgency	Priority/urgency of the case	Decimal	
Description	Description for this individual case instance	Text	✓
Created by	User that created the case	User Refere...	
Created date/time	Date/time case was created	DateTime	
Resolved by	User that resolved the case	User Refere...	
Resolved date/time	Date/time case was resolved	DateTime	
Vehicle Under Maint...	The specific vehicle undergoing maintenance, inspection, or repair.	(Data)	
Maintenance Location	The location where the fleet maintenance activities are being performed.	(Data)	
Maintenance Workfo...	The employees assigned to perform the maintenance tasks or manage the fleet maintena...	(Data)	
Maintenance Task	Individual tasks associated with the inspection, servicing, or repair of vehicles.	(Data)	
Vehicle in Returns Q...	A record of a vehicle that has been returned by a customer and is awaiting inspection or m...	Text	
Vehicle in Dirty Car B...	A record of a vehicle that requires cleaning and/or maintenance before it can be rented ag...	(Data)	
Maintenance Overti...	Overtime requests or approvals related to workforce adjustments for maintenance activiti...	(Data)	

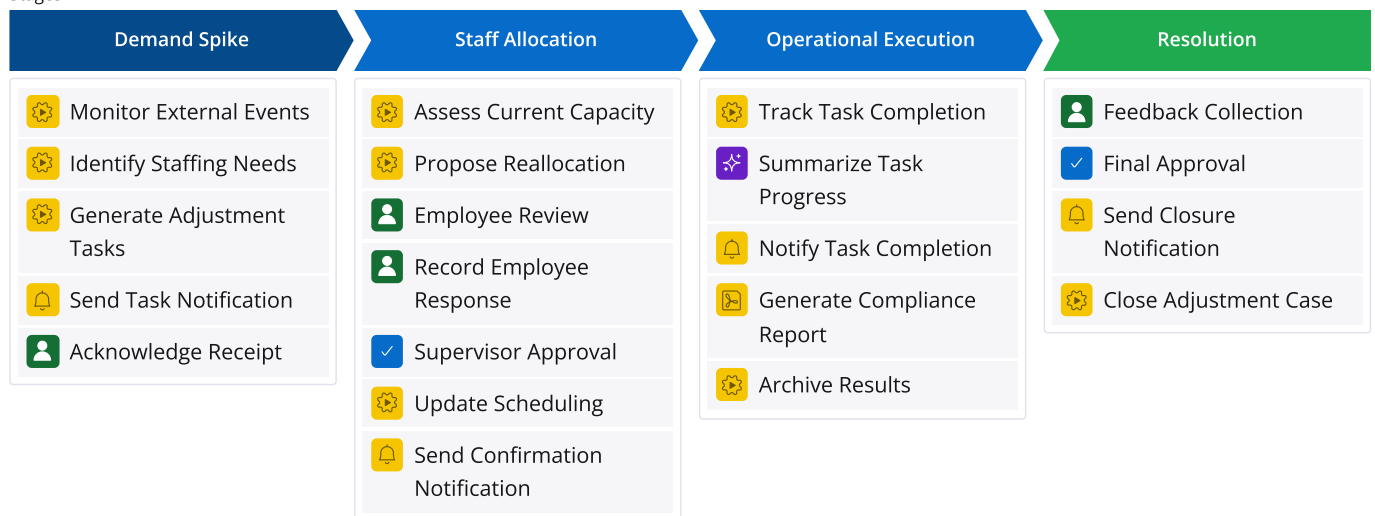
Business Rules

Name	Description
Maintenance Schedu...	Assesses vehicle service requirements, operational downtime impact, and maintenance resource availability to schedule ...
Decide Urgency & Pa...	Decision routing (In-Lane/Offsite/Defer) based on impact and resources.
Identify Open Issues	Decision: Are there unresolved problems or missed steps?
Detect Rework Trigger	Decision: Has work been rejected or has missing/incorrect info been found?
Supervisor Decision	Management determines immediate actions, e.g., hold, external service referral, or reporting.

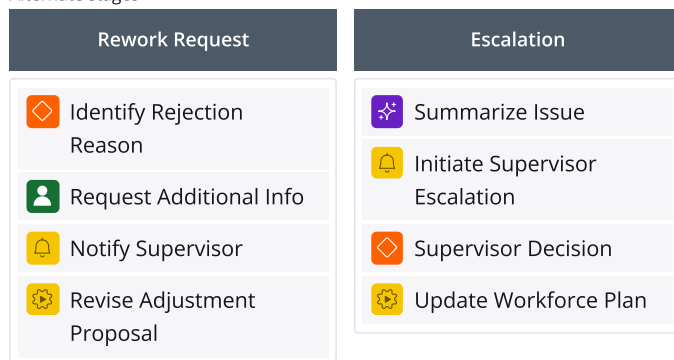
Staffing Adjustment

Facilitates human-in-the-loop workforce optimization such as reallocation or overtime requests, reacting dynamically to demand spikes, staffing shortages, and operational disruptions. Employees receive actionable notifications and respond directly through the mobile app.

Stages



Alternate stages



Case Data Model

Field Name	Description	Type	Primary
Staffing adjustment ...	Business title for the specific staffing adjustment instance initiated to address workforce ne...	Text	✓
Reason for adjustme...	Business rationale for the workforce adjustment, such as demand spike, staffing shortage, ...	Text	✓
Action type	Type of staffing intervention, such as reallocation or overtime.	Text	✓
Adjustment status	Current state of the staffing adjustment (e.g., suggested, pending approval, accepted, reject...	Text	
Operational area	The affected area of operations, such as returns, cleaning, or front desk, where the adjust...	Text	
Proposed start time	Date and time when the staffing adjustment is suggested to begin.	DateTime	
Proposed end time	Date and time when the staffing adjustment is suggested to end.	DateTime	
Location	The Hertz branch, depot, or airport facility where the staffing adjustment applies.	(Data)	
Impacted employee	The employee affected by or targeted for the staffing adjustment.	(Data)	
Suggested by AI agent	Name of the AI agent or system module that recommended the adjustment.	Text	
Relevant weather ev...	Weather-related event influencing the need for the staffing adjustment.	(Data)	
Related flight	Flight associated with the staffing adjustment, typically for airport operations.	(Data)	
Linked task	Operational task associated with the staffing adjustment, such as a vehicle move request o...	(Data)	
Customer demand tr...	Demand event or metric that triggered the staffing adjustment, such as an anticipated spik...	Text	
Employee response	Response from the employee (accept or reject) to the adjustment notification in the mobile ...	Text	

Field Name	Description	Type	Primary
Case Label	Label for this individual case instance	Text	✓
Case ID	Unique identifier for this individual case instance	Text	✓
Work Status	Work status of the case	Text	✓
Urgency	Priority/urgency of the case	Decimal	
Description	Description for this individual case instance	Text	✓
Created by	User that created the case	User Refer...	
Created date/time	Date/time case was created	DateTime	
Resolved by	User that resolved the case	User Refer...	
Resolved date/time	Date/time case was resolved	DateTime	
Affected Location	The specific location where the staffing adjustment is needed or being applied.	(Data)	
Affected Employees	The employees whose staffing, roles, or schedules are being adjusted.	(Data)	
Triggering Weather E...	The weather event that caused operational disruption and necessitated staffing adjustments.	(Data)	
Related Flights	Flights contributing to demand spikes or operational disruptions requiring staffing adjustm...	(Data)	
Associated Returns ...	The returns queue whose status (e.g., backlog) is influencing staffing adjustment decisions.	(Data)	
Associated Dirty Car ...	The dirty car backlog whose size is influencing staffing adjustment decisions.	(Data)	
Related Overtime Re...	Overtime requests generated or approved as part of this staffing adjustment.	(Data)	

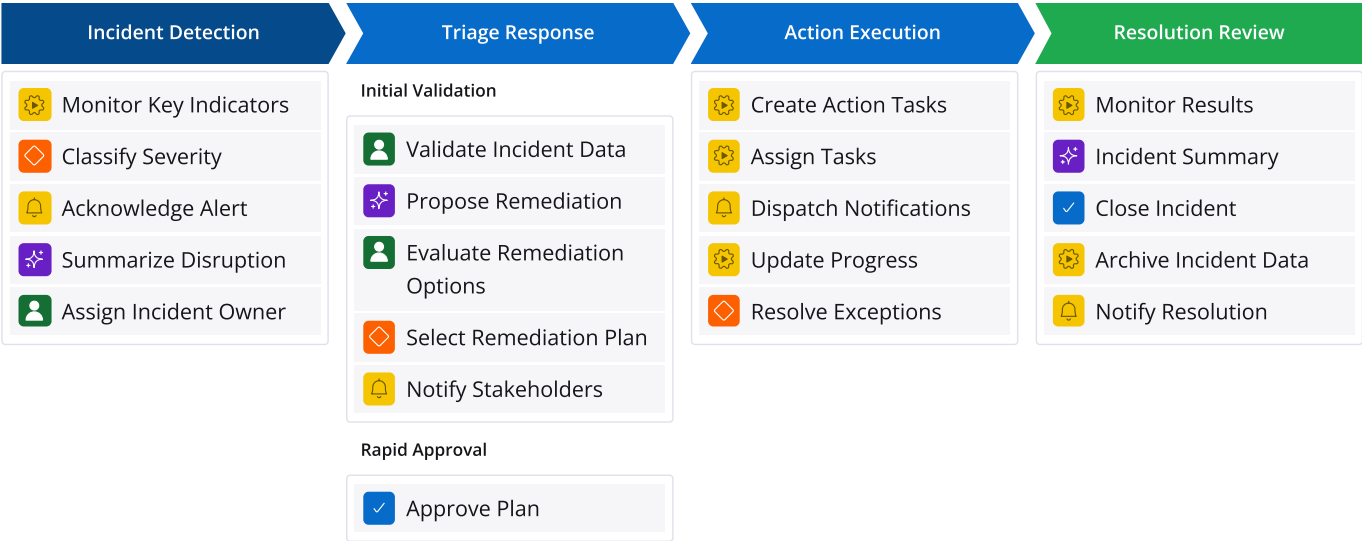
Business Rules

Name	Description
Workforce Reallocati...	Analyzes staffing levels against operational demand, backlog indicators, and location-specific workload to determine whe...
Identify Rejection Re...	Analyze reason for rejection or missing information.
Supervisor Decision	Supervisor reviews and decides on adjustment outcome.

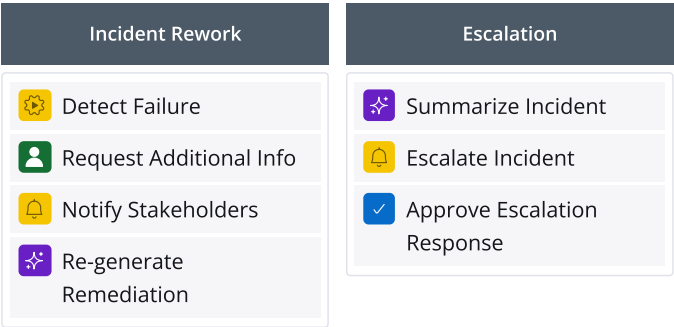
Operational Incident

Handles the identification, escalation, and resolution of operational disruptions—such as weather, flight delays, or sudden customer demand changes. AI agents monitor conditions and automate remediation proposals to maintain service continuity.

Stages



Alternate stages



Case Data Model

Field Name	Description	Type	Primary ⓘ
Operational incident...	Business-friendly name/title identifying the operational incident instance.	Text	✓
Incident type	Type of operational disruption affecting fleet operations. Examples include weather event, fli...	Text	✓
Incident description	Summary detailing the nature, cause, and business impact of the operational incident.	Text	✓
Incident date/time	Date and time when the operational incident was detected or reported.	DateTime	
Detected by AI agent	Identification of the AI agent or system component that detected the incident.	Text	
Proposed remediat...	Suggested actions generated by the AI agent for resolving or mitigating the incident, such as ...	Text	
Remediation action ...	Status indicating whether employees have accepted or rejected proposed remediation actio...	Text	
Escalation required	Indicator showing if the incident needs escalation to a manager/lead for manual intervention.	Boolean	
Escalated to	User reference for the manager or supervisor to whom the incident is escalated.	User Refe...	
Operational status in...	Status indicator describing the operational state related to this incident (e.g., red for backlog...	Text	
Incident outcome	Result of incident resolution, specifying if the disruption was managed, mitigated, or unresol...	Text	
Affected location	Location where the operational incident has primary impact, such as airport, branch, or dep...	(Data)	
Affected vehicle(s)	Vehicles directly involved in or impacted by the operational incident.	(Data)	
Affected employees	Employees impacted by the incident, such as transporters or vehicle service associates.	(Data)	
Impacted customer(s)	Customers affected by operational disruption, such as delays in vehicle pickup or return.	(Data)	

Field Name	Description	Type	Primary
Related tasks	Tasks generated in response to the incident, such as vehicle movement, cleaning, or workfor...	(Data)	
Related weather event	Weather event related to or triggering the operational incident.	(Data)	
Related flight	Flight event triggering or related to the incident, such as delays impacting operations.	(Data)	
Dirty car backlog im...	Indicator describing how the incident affects the backlog of cars needing cleaning or mainte...	(Data)	
Case Label	Label for this individual case instance	Text	✓
Case ID	Unique identifier for this individual case instance	Text	✓
Work Status	Work status of the case	Text	✓
Urgency	Priority/urgency of the case	Decimal	
Description	Description for this individual case instance	Text	✓
Created by	User that created the case	User Refe...	
Created date/time	Date/time case was created	DateTime	
Resolved by	User that resolved the case	User Refe...	
Resolved date/time	Date/time case was resolved	DateTime	
Incident-Impacted V...	The specific vehicle impacted or involved in the operational incident, requiring assessment o...	(Data)	
Incident Location	The physical location where the operational incident occurred or is primarily impacting oper...	(Data)	
Affected Customer	A customer whose rental plans or experience is directly impacted by the operational incident.	(Data)	
Responding Employee	An employee assigned to manage, investigate, or resolve aspects of the operational incident.	(Data)	
Incident Remediatio...	A specific action or task generated to address or mitigate the impact of the operational incid...	(Data)	
Triggering Weather E...	The specific weather event that initiated or significantly contributed to the operational incide...	(Data)	
Impacted Flight	A flight whose delay or cancellation is directly causing or contributing to the operational inci...	(Data)	
Affected Returns Qu...	The vehicle returns queue at a specific location that is experiencing disruption due to the op...	Text	
Incident-Related Dirt...	A backlog of dirty vehicles requiring cleaning, directly resulting from or exacerbated by the o...	(Data)	
Incident-Related Ove...	An overtime request generated to address staffing needs for resolving or mitigating the ope...	(Data)	

Business Rules

Name	Description
Disruption Impact As...	Evaluates external factors including weather events, flight delays, and demand fluctuations against current operational c...
AI Agent Alert Trigge...	Monitors predefined operational thresholds and external data sources to automatically trigger AI agent alerts for airport ...
Classify Severity	Assess impact level based on metrics and status indicators.
Select Remediation ...	Choose the most appropriate plan based on business context.
Resolve Exceptions	Flag and route exceptions for manual review as needed.

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